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Electronic package and manufacturing method thereof

Abstract

An electronic package is provided, in which an electronic element is disposed on an upper side of a circuit structure, a package layer covers the electronic element, and an action structure is embedded in the package layer, so that the action structure is exposed from a surface of the package layer, and then a bonding element is disposed on a lower side of the circuit structure and corresponding to the position of the action structure, so as to form a thermal conduction between the bonding element and the action structure. Therefore, a laser can transfer heat energy to the bonding element via the action structure, so that a solder material on the bonding element can be reflowed.

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Background/Summary

BACKGROUND

1. Technical Field

(1) The present disclosure relates to a semiconductor packaging process, and more particularly, to an electronic package with a heat dissipation mechanism and a manufacturing method thereof.

2. Description of Related Art

(2) With the evolution of technology, the demand trend of electronic products is moving towards high-end products with high-density lines/high transmission speed/high number of stacked layers/large-size designs. These products are more sensitive to thermal reactions as the chip size increases and the number of contacts (I/O) increases. Therefore, the thermal process (such as the reflow process) in the packaging operation is very likely to cause warpage of the overall structure due to different coefficients of thermal expansion (CTE) of various materials, and the problem of poor reliability would also occur due to the concentration of thermal stress inside the structure.

(3) At present, the laser assisted bonding (LAB) process may selectively perform a local heating, and has the characteristics of rapid temperature rise. Therefore, the thermal process time can be greatly reduced, thereby reducing the thermal stress concentration inside the structure, and by controlling the laser wavelength and the characteristics of local heating, the degree of warpage can be greatly reduced.

(4) FIG. 1 is a schematic view of a conventional semiconductor package 1. As shown in FIG. 1, the semiconductor package 1 is provided with a semiconductor chip 11 that is disposed on a substrate structure 10 having a dielectric layer 100 and a routing layer 101 in a flip-chip manner (via solder bumps 13), and the semiconductor chip 11 is then covered with a package layer 12. Afterwards, conductive bumps 14, 15 on the lower side of the substrate structure 10 can connect a plurality of solder materials 16, 17 onto contacts 19 of a circuit board 1a via the LAB process.

(5) However, during the LAB process, the thermal energy of a laser L can only penetrate through the semiconductor chip 11 but cannot penetrate through the package layer 12, resulting in insufficient thermal energy of the conductive bumps 14 under the package layer 12, and thus causing the problem of non-wetting of the solder materials 16 there.

(6) Therefore, how to overcome the above-mentioned drawbacks of the prior art has become an urgent issue to be solved at present.

SUMMARY

(7) In view of the various deficiencies of the prior art, the present disclosure provides an electronic package, which comprises: a circuit structure having a first side and a second side opposing the first side; an electronic element disposed on the first side of the circuit structure; a package layer disposed on the first side of the circuit structure to cover the electronic element; at least one action structure embedded in the package layer and located around the electronic element, wherein the action structure is exposed from an upper surface of the package layer and connected to the first side of the circuit structure; and at least one bonding element disposed on the second side of the circuit structure and corresponding to a position of the action structure, wherein a thermal conduction is formed between the bonding element and the action structure.

(8) The present disclosure further provides a method of manufacturing an electronic package, the method comprises: providing a circuit structure having a first side and a second side opposing the first side; disposing an electronic element on the first side of the circuit structure; forming a package layer on the first side of the circuit structure to cover the electronic element, and embedding at least one action structure in the package layer around the electronic element, wherein the action structure is exposed from an upper surface of the package layer and connected to the first side of the circuit structure; and forming at least one bonding element on the second side of the circuit structure, wherein the bonding element corresponds to a position of the action structure, such that a thermal conduction is formed between the bonding element and the action structure.

(9) In the aforementioned method, the action structure is first formed on the first side of the circuit structure, and then the package layer covers the action structure and the electronic element together.

(10) In the aforementioned method, after disposing the electronic element on the first side of the circuit structure, the package layer is made to cover the electronic element, and then a through via exposing the first side is formed on the package layer, and then a metal material is filled in the through via to form the action structure.

(11) In the aforementioned electronic package and the manufacturing method thereof, the action structure includes a pillar free from having electrical function.

(12) In the aforementioned electronic package and the manufacturing method thereof, the action structure is connected to a functional part of the circuit structure.

(13) In the aforementioned electronic package and the manufacturing method thereof, the circuit structure includes a fan-out redistribution layer.

(14) In the aforementioned electronic package and the manufacturing method thereof, the action structure includes a pillar, and the pillar is used as a center and forms a circular heat-affected zone by two to three times a radius of the pillar, wherein the heat-affected zone is projected vertically in a direction from the first side to the second side to define a heat channel in the circuit structure, such that the bonding element at least partially and correspondingly falls within a range of the heat channel.

(15) In the aforementioned electronic package and the manufacturing method thereof, the action

structure includes a plurality of pillars, such that the single bonding element corresponds to at least two of the pillars.

(16) In the aforementioned electronic package and the manufacturing method thereof, the action structure includes at least one pillar, such that the single pillar corresponds to a plurality of the bonding elements.

(17) In the aforementioned electronic package and the manufacturing method thereof, the action structure includes at least one pillar, and a ratio of a diameter of the pillar to a diameter of the bonding element is 0.2 to 0.4 or 0.8 to 1.2.

(18) As can be seen from the above, in the electronic package and the manufacturing method thereof of the present disclosure, the action structure is embedded in the package layer and exposed from the surface of the package layer, so that the laser can transfer heat energy to the bonding element on the second side of the circuit structure via the action structure, such that the solder material on the bonding element can be reflowed. Therefore, compared with the prior art, the present disclosure can effectively improve the problem of non-wetting of the solder material on the bonding element.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic cross-sectional view of a conventional semiconductor package.

(2) FIG. 2A to FIG. 2D are schematic cross-sectional views illustrating a manufacturing method of an electronic package according to the present disclosure.

(3) FIG. 2E is a schematic cross-sectional view of the subsequent process of FIG. 2D.

(4) FIG. 3A to FIG. 3D are schematic partial bottom views of FIG. 2D.

(5) FIG. 4A is a schematic cross-sectional view illustrating another method of FIG. 2B to FIG. 2C.

(6) FIG. 4B is a schematic cross-sectional view illustrating another method of the subsequent process of FIG. 2D.

DETAILED DESCRIPTIONS

(7) The following describes the implementation of the present disclosure with examples. Those skilled in the art can easily understand other advantages and effects of the present disclosure from the contents disclosed in this specification.

(8) It should be understood that, the structures, ratios, sizes, and the like in the accompanying figures are used for illustrative purposes to facilitate the perusal and comprehension of the contents disclosed in the present specification by one skilled in the art, rather than to limit the conditions for practicing the present disclosure. Any modification of the structures, alteration of the ratio relationships, or adjustment of the sizes without affecting the possible effects and achievable proposes should still be deemed as falling within the scope defined by the technical contents disclosed in the present specification. Meanwhile, terms such as “upper,” “lower,” “first,” “second,” “a,” “one” and the like used herein are merely used for clear explanation rather than limiting the practicable scope of the present disclosure, and thus, alterations or adjustments of the relative relationships thereof without essentially altering the technical contents should still be considered in the practicable scope of the present disclosure.

(9) FIG. 2A to FIG. 2D are schematic cross-sectional views illustrating a manufacturing method of an electronic package 2 of the present disclosure.

(10) As shown in FIG. 2A, a circuit structure 20 is arranged on a support board 9, and an action structure 2a including a plurality of first pillars 23 or including a first pillar 23 and a second pillar 28 is formed on the circuit structure 20.

(11) In one embodiment, the support board 9 is, for example, a board body made of semiconductor material (e.g., silicon or glass), and the circuit structure 20 is, for example, a package substrate with

a core layer, a coreless package substrate, a through silicon interposer (TSI) with a through-silicon via (TSV), or other board types, wherein the circuit structure **20** has a first side **20a** and a second side **20b** opposite to the first side **20a**, so that the circuit structure **20** is bonded onto the support board **9** with the second side **20b** thereof. For example, the circuit structure **20** includes at least one insulating layer **200** and at least one routing layer **201** bonded with the insulating layer **200**, such as at least one fan-out type redistribution layer (RDL), and the circuit structure **20** further has at least one functional part **202**, such as for signal transmission or for grounding. It should be understood that the circuit structure **20** can also be other substrates for carrying chips, such as lead frames, wafers, or other board bodies with metal routings, etc., which are not limited to the above.

(12) Furthermore, the plurality of first pillars **23** and the second pillar **28** are erected or vertically disposed on the first side **20a** of the circuit structure **20**, and the second pillar **28** can be connected to the functional part **202** of the circuit structure **20** according to requirements. For example, the material for forming the first and second pillars **23**, **28** is a metal material such as copper, a solder material, or other materials that are easy to conduct heat.

(13) In addition, the routing method of the circuit structure **20** is a fan-out redistribution layer (FORDL), so a pitch of the routing layer **201** fans out from a smaller pitch of the first side **20a** to a larger pitch of the second side **20b**, that is, the pitch of the routing layer **201** on the first side **20a** is smaller than the pitch of the routing layer **201** on the second side **20b**.

(14) As shown in FIG. 2B, at least one electronic element **21** is disposed on the first side **20a** of the circuit structure **20**, and the first pillars **23** are free from being electrically connected to the electronic element **21** to serve as dummy vias.

(15) In one embodiment, the electronic element **21** is an active element, a passive element or a combination of the active element and the passive element, etc., wherein the active element is, for example, a semiconductor chip, and the passive element is, for example, a resistor, a capacitor, or an inductor. In one embodiment, the electronic element **21** is a semiconductor chip, which has an active surface **21a** and an inactive surface **21b** opposite to the active surface **21a**, wherein the active surface **21a** has a plurality of electrode pads, so as to be disposed on the first side **20a** of the circuit structure **20** in a flip chip manner by a plurality of conductive bumps **211** such as solder material, metal pillars or the like and electrically connected to the routing layer **201** with the smaller pitch, so that these high-density electrode pads or conductive bumps **211** can be fanned out via the circuit structure **20** to contacts on the second side **20b** with the larger pitch, and the conductive bumps **211** are covered with an insulating material **212** such as an underfill or a non-conductive underfill thin film (NCF); alternatively, the electronic element **21** can be electrically connected to the routing layer **201** of the circuit structure **20** via a plurality of bonding wires (not shown) in a wire bonding manner; or the electronic element **21** may directly contact the routing layer **201** of the circuit structure **20**. Therefore, there are many ways in which the electronic element **21** is electrically connected to the circuit structure **20**, which are not limited to the above.

(16) Furthermore, the first and second pillars **23**, **28** are located around the electronic element **21**, such as surrounding the electronic element **21**, with the first and second pillars **23**, **28** as the center and with two to three times its radius D forming a circular heat-affected zone **A**, so that the heat-affected zone **A** is projected vertically in a direction from the first side **20a** to the second side **20b**, so as to define a cylindrical heat channel **S** in the circuit structure **20**. It can be understood that the size of the circular heat-affected zone **A** varies with the thermal diffusivity of the material of the first and second pillars **23**, **28**. In an embodiment, the first and second pillars **23**, **28** are made from copper, but the present disclosure is not limited to as such.

(17) It should be understood that, in other embodiments, the electronic element **21** can also be disposed on the circuit structure **20** first, and then the first and second pillars **23**, **28** are formed.

(18) As shown in FIG. 2C, a package layer **22** is formed on the first side **20a** of the circuit structure **20**, so that the package layer **22** covers the electronic element **21** and the circuit structure **20** together, such that part of the surface (such as the upper surfaces of the first and second pillars **23**,

(28) of the action structure **2a** is exposed from the package layer **22**.

(19) In one embodiment, the package layer **22** is an insulating material, such as polyimide (PI), dry film, encapsulant such as epoxy resin, or molding compound.

(20) Furthermore, a surface **22a** of the package layer **22** can be flushed with the inactive surface **21b** of the electronic element **21** and end surfaces **23a** and **28a** of the first and second pillars **23** and **28** via a leveling process, so that the inactive surface **21b** of the electronic element **21** and the end surfaces **23a** and **28a** of the first and second pillars **23** and **28** are exposed from the surface **22a** of the package layer **22**. For example, the leveling process removes part of the material of the electronic element **21**, part of the material of the action structure **2a** and part of the material of the package layer **22** by grinding. It should be understood that, the package layer **22** can also expose the inactive surface **21b** of the electronic element **21** and the end surfaces **23a** and **28a** of the first and second pillars **23** and **28** in the form of openings.

(21) Also, in another embodiment, as shown in FIG. 4A, after the electronic element **21** is disposed on the circuit structure **20**, the package layer **22** may be formed first, and then at least one or a plurality of through vias **40** exposing the insulating layer **200** (even exposing the functional part **202**) are formed on the package layer **22**. Afterwards, an easily thermally conductive material is formed in the through vias **40** to serve as the action structure **2a**.

(22) As shown in FIG. 2D, the support board **9** is removed to expose the second side **20b** of the circuit structure **20**, and a singulation process is performed along a cutting path Y shown in FIG. 2C. Next, a plurality of bonding elements **24** and a plurality of conductive elements **25** are formed on the second side **20b** of the circuit structure **20** to form the electronic package **2** of the present disclosure, wherein the bonding elements **24** and the conductive elements **25** are electrically connected to the routing layer **201**, so that the electronic element **21** is electrically connected to the conductive elements **25** and the bonding elements **24**.

(23) In one embodiment, positions of the first and second pillars **23**, **28** and the bonding elements **24** are respectively disposed on the first side **20a** and the second side **20b** of the circuit structure **20**, so that the positions of the first and second pillars **23**, **28** and the bonding elements **24** are arranged to be correspond to each other up and down.

(24) Furthermore, the bonding elements **24** and the conductive elements **25** are metal bumps such as copper or other conductive materials, and the materials of the bonding elements **24**, the conductive elements **25** and the conductive bumps **211** can be the same or different.

(25) In addition, the bonding element **24** is correspondingly at least partially within the range of the heat channel S defined by the circular heat-affected zone A. For example, the bonding element **24** entirely falls into the range of the heat channel S in a concentric manner relative to the first pillar **23**, as shown in FIG. 3A; or the bonding element **24** entirely falls into the range of the heat channel S in an eccentric manner relative to the first pillar **23**, as shown in FIG. 3B. In other words, the relative positions of the first pillar **23** and the bonding element **24** can be in a center-aligned relationship (as shown in FIG. 3A) or a non-aligned relationship (as shown in FIG. 3B), as long as the position of the bonding element **24** intersects the heat channel S will do.

(26) It should be understood that there are many ways for the bonding element **24** to fall into the heat channel S. As shown in FIG. 3C, the position of the bonding element **24** partially intersects the heat channel S (four circular heat-affected zones A1, A2, A3, A4), and the present disclosure is not limited to as such.

(27) In addition, a single bonding element **24** may correspond to a plurality of pillars, such as the four first pillars **23** shown in FIG. 3C, and the first pillars **23** are symmetrically arranged relative to the bonding element **24**, so that the bonding element **24** only partially falls within the range of the heat channel S defined by the four heat-affected zones A1, A2, A3, and A4, wherein the ratio ($R1/R$ or $R2/R$) of the diameter R1 of the first pillar **23** (or the diameter R2 of the second pillar **28** as shown in FIG. 2D) to the diameter R of the bonding element **24** may be between 0.2 and 0.4. Alternatively, a single pillar corresponds to a plurality of the bonding elements **24**, as shown in

FIG. 3D, one first pillar **23** corresponds to four bonding elements **24**, and the bonding elements **24** are symmetrically arranged relative to the first pillar **23** and fall within the range of the heat channel **S** completely correspondingly, wherein the ratio ($R1/R$ or $R2/R$) of the diameter $R1$ of the first pillar **23** (or the diameter $R2$ of the second pillar **28** as shown in FIG. 2D) to the diameter R of the bonding element **24** may be between 0.8 and 1.2. It should also be noted that the aforementioned FIG. 3A to FIG. 3D use the first pillar **23** as an illustration, and the bonding element **24** can also at least partially fall into the heat channel **S** defined by the circular heat-affected zone **A** of the second pillar **28**, so details are not repeated here.

(28) It should be understood that, the number and position of the first pillar **23** and the bonding element **24** can be adjusted with respect to the distribution of the heat-affected zones **A**, **A1**, **A2**, **A3**, **A4** according to requirements, and are not limited to the above.

(29) In the subsequent process, as shown in FIG. 2E, the electronic package **2** can be connected onto the contacts **30** of a circuit board **3a** with the bonding elements **24** and the conductive elements **25** thereof via the solder materials **26** and **27**, and then a reflow process is performed to assist heating and reflowing the solder materials **26**, **27** by laser **L**, wherein the laser **L** is irradiated from the package layer **22** toward the circuit structure **20**. Therefore, the laser **L** not only transmits heat energy to the second side **20b** of the circuit structure **20** by penetrating through the electronic element **21**, but also transmits heat energy to the second side **20b** of the circuit structure **20** via a heating path (i.e., from the action structure **2a** to the heat channel **S**), so that the heated and reflowed solder material **26** on the bonding element **24** and the heated and reflowed solder material **27** on the conductive element **25** can be strengthened. It should be understood that since the laser **L** cannot penetrate through the package layer **22**, the first and second pillars **23** and **28** of the action structure **2a** need to be exposed from the package layer **22** and penetrate through the package layer **22**, so that the first and second pillars **23**, **28** communicate from the package layer **22** to the circuit structure **20**.

(30) Alternatively, as shown in FIG. 4B, in the subsequent process, at least one package module **4a** can be stacked on the end surfaces of the first and second pillars **23**, **28** exposed from the package layer **22** via a solder material **46**, to form a stacked package **4** (Package on Package [Pa]), wherein the package module **4a** includes at least one semiconductor chip **41**, so the package module **4a** can be designed according to requirements, such as being similar to the aspect of the electronic package **2** in FIG. 2D, and the present disclosure is not limited to as such.

(31) Therefore, in the manufacturing method of the present disclosure, the action structure **2a** is embedded in the package layer **22** to serve as a heat conduction path for the laser assisted bonding (LAB) process, so that the laser **L** can pass through the package layer **22** via the first and second pillars **23**, **28** penetrating through the package layer **22** to heat the bonding element **24** under the heat channel **S**, and the positions of the first and second pillars **23** and **28** and the bonding elements **24** are corresponding to each other up and down, so that the reflowed solder material **26** on the bonding element **24** can be strengthened. Therefore, compared with the prior art, the manufacturing method of the present disclosure can effectively improve the problem of non-wetting of the solder material **26**, so that the solder material **26** on the bonding element **24** can be smoothly melted and fastened to the circuit board **3a**.

(32) Furthermore, by the position of the bonding element **24** falling within the range of the heat channel **S**, the heating performance of the LAB process is improved.

(33) Further, in the LAB process, if the circuit structure **20** is a coreless package substrate that is more prone to warping, the benefit of improving the thermal stress problem can be further highlighted by the design of the first pillar **23**.

(34) In addition, if the action structure **2a** (such as the second pillar **28**) is connected to the functional part **202** for grounding of the circuit structure **20**, in addition to serving as a heat conduction path for the LAB process, it can also provide a shielding function, and if the action structure **2a** (such as the second pillar **28**) is connected to the functional part **202** for transmitting

signals, it can also provide an electrical path for the upper and lower connection of the stacked package **4**.

(35) The present disclosure also provides an electronic package **2**, which comprises: a circuit structure **20**, at least one electronic element **21**, a package layer **22**, an action structure **2a** and at least one bonding element **24**.

(36) The circuit structure **20** has a first side **20a** and a second side **20b** opposite to the first side **20a**.

(37) The electronic element **21** is disposed on the first side **20a** of the circuit structure **20**.

(38) The package layer **22** is disposed on the first side **20a** of the circuit structure **20** to cover the electronic element **21**.

(39) The action structure **2a** is embedded in the package layer **22** and connected to the first side **20a** of the circuit structure **20** and located around the electronic element **21**, wherein the action structure **2a** is exposed from the surface **22a** of the package layer **22**.

(40) The bonding element **24** is disposed on the second side **20b** of the circuit structure **20** corresponding to the position of the action structure **2a**, so that thermal conduction is formed between the bonding element **24** and the action structure **2a**.

(41) In one embodiment, the action structure **2a** includes a first pillar **23** free from having electrical function.

(42) In one embodiment, the action structure **2a** includes a second pillar **28**, which is connected to the functional part **202** of the circuit structure **20**.

(43) In one embodiment, the circuit structure **20** includes a routing layer **201** such as a fan-out redistribution layer.

(44) In one embodiment, the first and second pillars **23**, **28** are used as the center and form a circular heat-affected zone A, A1, A2, A3, A4 by two to three times the radius D of the first and second pillars **23**, **28**, so that the heat-affected zone A, A1, A2, A3, and A4 is projected vertically in a direction from the first side **20a** to the second side **20b**, so as to define a heat channel S in the circuit structure **20**, so that the bonding element **24** at least partially and correspondingly falls within the range of the heat channel S.

(45) In one embodiment, a single bonding element **24** corresponds to a plurality of the pillars (at least including a plurality of first pillars **23**, and may include the second pillar **28** according to requirements).

(46) In one embodiment, the single first pillar **23** or the second pillar **28** corresponds to a plurality of the bonding elements **24**.

(47) In one embodiment, the diameter ratio ($R1/R$) of the first pillar **23** to the bonding element **24** is 0.2 to 0.4 or 0.8 to 1.2, and the diameter ratio ($R2/R$) of the second pillar **28** to the bonding element **24** is 0.2 to 0.4 or 0.8 to 1.2.

(48) To sum up, in the electronic package and the manufacturing method thereof of the present disclosure, the action structure is embedded in the package layer, so that the laser can pass through the package layer and heat the bonding element under the heat channel via the action structure, thereby transferring heat energy to the bonding element to reflow the solder material thereon. Therefore, the present disclosure can effectively improve the problem of non-wetting of the solder material.

(49) The foregoing embodiments are provided for the purpose of illustrating the principles and effects of the present disclosure, rather than limiting the present disclosure. Anyone skilled in the art can modify and alter the above embodiments without departing from the spirit and scope of the present disclosure. Therefore, the scope of protection with regard to the present disclosure should be as defined in the accompanying claims listed below.

Claims

1. An electronic package, comprising: a circuit structure having a first side and a second side opposing the first side; an electronic element disposed on the first side of the circuit structure; a package layer disposed on the first side of the circuit structure to cover the electronic element; at least one action structure embedded in the package layer and located around the electronic element, wherein the action structure is exposed from an upper surface of the package layer and connected to the first side of the circuit structure; and at least one bonding element disposed on the second side of the circuit structure and corresponding to a position of the action structure, wherein a thermal conduction is formed between the bonding element and the action structure, wherein the action structure includes a single pillar or a plurality of pillars, and wherein the at least one bonding element includes a single bonding element or a plurality of bonding elements, such that the single bonding element corresponds to at least two of the pillars or the single pillar corresponds to the plurality of the bonding elements.
2. The electronic package of claim 1, wherein a said pillar is free from having electrical function.
3. The electronic package of claim 1, wherein the action structure is connected to a functional part of the circuit structure.
4. The electronic package of claim 1, wherein the circuit structure includes a fan-out redistribution layer.
5. The electronic package of claim 1, wherein a said pillar is used as a center and forms a circular heat-affected zone by two to three times a radius of the pillar, and wherein the heat-affected zone is projected vertically in a direction from the first side to the second side to define a heat channel in the circuit structure, such that a said bonding element at least partially and correspondingly falls within a range of the heat channel.
6. The electronic package of claim 1, wherein a ratio of a diameter of a said pillar to a diameter of a said bonding element is 0.2 to 0.4 or 0.8 to 1.2.
7. A method of manufacturing an electronic package, comprising: providing a circuit structure having a first side and a second side opposing the first side; disposing an electronic element on the first side of the circuit structure; forming a package layer on the first side of the circuit structure to cover the electronic element, and embedding at least one action structure in the package layer around the electronic element, wherein the action structure is exposed from an upper surface of the package layer and connected to the first side of the circuit structure; and forming at least one bonding element on the second side of the circuit structure, wherein the at least one bonding element corresponds to a position of the action structure, such that a thermal conduction is formed between the at least one bonding element and the action structure, wherein the action structure includes a single pillar or a plurality of pillars, and wherein the at least one bonding element includes a single bonding element or a plurality of bonding elements, such that the single bonding element corresponds to at least two of the pillars or the single pillar corresponds to the plurality of the bonding elements.
8. The method of claim 7, wherein a said pillar is free from having electrical function.
9. The method of claim 7, wherein the action structure is connected to a functional part of the circuit structure.
10. The method of claim 7, wherein the circuit structure includes a fan-out redistribution layer.
11. The method of claim 7, wherein said pillar is used as a center and forms a circular heat-affected zone by two to three times a radius of the pillar, and wherein the heat-affected zone is projected vertically in a direction from the first side to the second side to define a heat channel in the circuit structure, such that a said bonding element at least partially and correspondingly falls within a range of the heat channel.
12. The method of claim 7, wherein a ratio of a diameter of a said pillar to a diameter of a said bonding element is 0.2 to 0.4 or 0.8 to 1.2.
13. The method of claim 7, wherein the action structure is first formed on the first side of the

circuit structure, and then the package layer covers the action structure and the electronic element together.

14. The method of claim 7, wherein after disposing the electronic element on the first side of the circuit structure, the package layer is made to cover the electronic element, and then at least one through via exposing the first side is formed on the package layer, and then a metal material is filled in the at least one through via to form the action structure.
